

Supplementary Appendix

TreeFlow: Probabilistic Modelling and Automatic Differentiation for Phylogenetics

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Model specification

TreeFlow provides two ways to specify phylogenetic models. For standard phylogenetic analyses, models can be defined using a YAML-based model definition format that is used as input to TreeFlow’s command line interfaces. For more flexible model specification, the Python developer API allows users to directly compose TensorFlow Probability distributions into a joint distribution object.

YAML model definition format

The YAML model definition format provides a concise, text-based specification of standard phylogenetic models. The format is a nested dictionary with keys at various levels corresponding to model components (tree prior, substitution model, clock model, site model), their parameters, and the prior distributions on those parameters. Figure S1 shows an example YAML model definition for a phylogenetic analysis of influenza sequences, specifying a coalescent tree prior, strict molecular clock, discretized Gamma site rate model, and GTR substitution model with independent Gamma priors on the relative substitution rates.

Python API model specification

For models that go beyond the standard form supported by the YAML format, the Python developer API provides direct access to TensorFlow Probability’s joint distribution framework. This allows users to compose any combination of distributions and transformations into a probabilistic graphical model. Figure S2 shows an example of this flexibility: the code specifies a phylogenetic model for the carnivores dataset, with highlighted lines showing the small changes required to extend the model from a single transition-transversion ratio (κ) to a per-lineage κ parameter. This extension, which leverages TensorFlow’s vectorization and broadcasting functionality, would be difficult to express in a fixed model definition format.

For further examples and a detailed tutorial on using TreeFlow’s Python API and command line interfaces, see the TreeFlow documentation at <https://treeflow.readthedocs.io>.

Choice of variational approximation family

TreeFlow provides three variational approximation families for fixed-topology models. The *mean field* family treats every coordinate of the (unconstrained) parameter vector as independent. The *full rank* family places a full covariance over the whole parameter vector, including every node height. The *root full rank* family, used for the analyses in the main text, places a full covariance over the substitution, clock and tree-prior parameters together with the tree’s root height, and treats the remaining node-height ratios as independent.

The motivation for the third family is that the two conventional choices fail in opposite directions on these models. The strongest posterior correlations in a time-tree analysis are between the parameters that jointly determine the scale of the tree: the clock rate, the tree prior’s rate parameter and the root height. A mean-field approximation cannot represent them at all, so it underestimates the marginal variance of exactly those quantities. A full-rank approximation can represent them, but it must also fit a covariance over all $n - 1$ node heights, which is $O(n^2)$ parameters for a problem whose useful correlation structure is concentrated in a handful of coordinates; in practice its optimization is slower to settle, its fitted marginal for the tree height is both biased and over-dispersed, and its clock rate is displaced. Restricting the covariance block to the shared parameters plus the root height keeps the correlations that matter while leaving the number of variational parameters linear in the number of taxa.

Figures S3 and S4 show the fitted posterior marginals for one run of each family on the carnivores and influenza datasets respectively, against the same BEAST 2 reference posterior. Tables S1 and S2 give the corresponding evidence lower bound and wall-clock runtime. ELBO values are comparable between approximations fitted to the same dataset, but not between datasets: the log likelihood is a sum over alignment patterns, of which the carnivores alignment has substantially more than the influenza alignment despite having far fewer taxa.

The bound is a poor guide to which family to prefer here. The mean field family is a special case of the root full rank family, one with a diagonal covariance block, so at their respective optima the latter cannot achieve a worse bound; but under a fixed optimization budget the two are separated on the influenza dataset by less than 0.02% of the ELBO, which is of the same order as the variation between independent runs of a single family. The marginals in Figure S4 are visibly different over the same pair of fits. The full rank family makes the point more sharply still. On the influenza dataset it is clearly separated on the bound, by more than a hundred times the gap between the other two families, and its marginals for the tree height, tree length and clock rate are displaced accordingly. On the carnivores dataset its bound is within 0.01% of the other two, yet its tree height and tree length marginals are grossly over-dispersed (Figure S3). Since the full rank family contains the other two, neither result reflects a lack of expressiveness; both are consequences of its much larger parameter count being harder to fit in the same number of iterations. A difference in the bound too small to separate the families can therefore accompany a large difference in the fitted posterior, and we would not recommend selecting between variational families for these models on the ELBO alone.

Approximation	ELBO	Runtime (minutes)
Mean field	-193,601	14.4
Full rank	-193,611	13.7
Root full rank	-193,600	13.9

Table S1: Evidence lower bound and wall-clock runtime for each variational approximation family on the carnivores dataset, for the runs shown in Figure S3. Each run is 60,000 optimization iterations.

Approximation	ELBO	Runtime (minutes)
Mean field	-21,561	59.3
Full rank	-21,815	66.0
Root full rank	-21,563	60.1

Table S2: Evidence lower bound and wall-clock runtime for each variational approximation family on the influenza (H3N2) dataset, for the runs shown in Figure S4. Each run is 60,000 optimization iterations.

Parameters that grow with the tree

The root full rank family confines its covariance block to parameters that are shared across the whole tree. A model may also have parameters that are replicated per branch, whose dimension grows with the tree in the same way the node heights do. Examples are an uncorrelated relaxed clock and the per-lineage transition-transversion ratio of the carnivores model in Figure S2. Including these in the covariance block would restore the $O(n^2)$ cost the family exists to avoid, so TreeFlow’s approximation constructor takes a `mean_field_vars` argument naming the variables to exclude from the block; they are then approximated independently, alongside the node-height ratios. Any posterior correlation between such a parameter and the rest of the model, for instance between a branch’s rate and the heights of the nodes it separates, is therefore not captured by this family, and remains a target for future work.

```

clock:
  strict:
    clock_rate:
      lognormal:
        loc: -2.0
        scale: 2.0
site:
  discrete_gamma:
    category_count: 4
    site_gamma_shape:
      lognormal:
        loc: 0.0
        scale: 1.0
substitution:
  gtr_rel:
    frequencies:
      dirichlet:
        concentration:
          - 2.0
          - 2.0
          - 2.0
          - 2.0
    rate_ac:
      gamma:
        concentration: 0.05
        rate: 0.05
    rate_ag:
      gamma:
        concentration: 0.05
        rate: 0.05
    rate_at:
      gamma:
        concentration: 0.05
        rate: 0.05
    rate_cg:
      gamma:
        concentration: 0.05
        rate: 0.05
    rate_gt:
      gamma:
        concentration: 0.05
        rate: 0.05
tree:
  coalescent:
    pop_size:
      lognormal:
        loc: 1.0
        scale: 1.5

```

Figure S1: YAML model definition for the analysis of the influenza dataset. The top-level keys specify the clock model, site rate model, substitution model, and tree prior. Each model component's parameters are given prior distributions from the standard distributions available in TensorFlow Probability.

```

site_category_count = 4
pattern_counts = alignment.get_weights_tensor()
subst_model = HKY()

def build_sequence_dist(tree, kappa, frequencies, site_gamma_shape):
    unrooted_tree = tree.get_unrooted_tree()
    site_rate_distribution = DiscretizedDistribution(
        category_count=site_category_count,
        distribution=Gamma(
            concentration=site_gamma_shape,
            rate=site_gamma_shape
        ),
    )
    transition_probs_tree = get_transition_probabilities_tree(
        unrooted_tree,
        subst_model,
        rate_categories=site_rate_distribution.normalised_support,
        frequencies=frequencies,
        frequencies=tf.broadcast_to(
            tf.expand_dims(frequencies, -2),
            kappa.shape + (4,)
        ),
    ),
    kappa=kappa
    )
    return SampleWeighted(
        DiscreteParameterMixture(
            site_rate_distribution,
            LeafCTMC(
                transition_probs_tree,
                expand_dims(frequencies, -2),
            ),
        ),
        sample_shape=alignment.site_count,
        weights=pattern_counts
    )

model = JointDistributionNamed(dict(
    birth_rate=LogNormal(c(1.0), c(1.5)),
    tree=lambda birth_rate: Yule(tree.taxon_count, birth_rate),
    kappa=LogNormal(c(0.0), c(2.0)),
    kappa=Sample(
        LogNormal(c(0.0), c(2.0)),
        tree.branch_lengths.shape
    ),
    site_gamma_shape=LogNormal(c(0.0), c(1.0)),
    frequencies=Dirichlet(c([2.0, 2.0, 2.0, 2.0])),
    sequences=build_sequence_dist
))

```

Figure S2: Code to specify models for the carnivores analysis using TreeFlow's Python API. Highlighted lines show changes (from previous line) to add kappa variation across lineages. The `Sample` distribution is used to make the kappa prior an independent identically-distributed sample for each branch. The `frequencies` parameter needs to be manually broadcasted over the added kappa dimension.

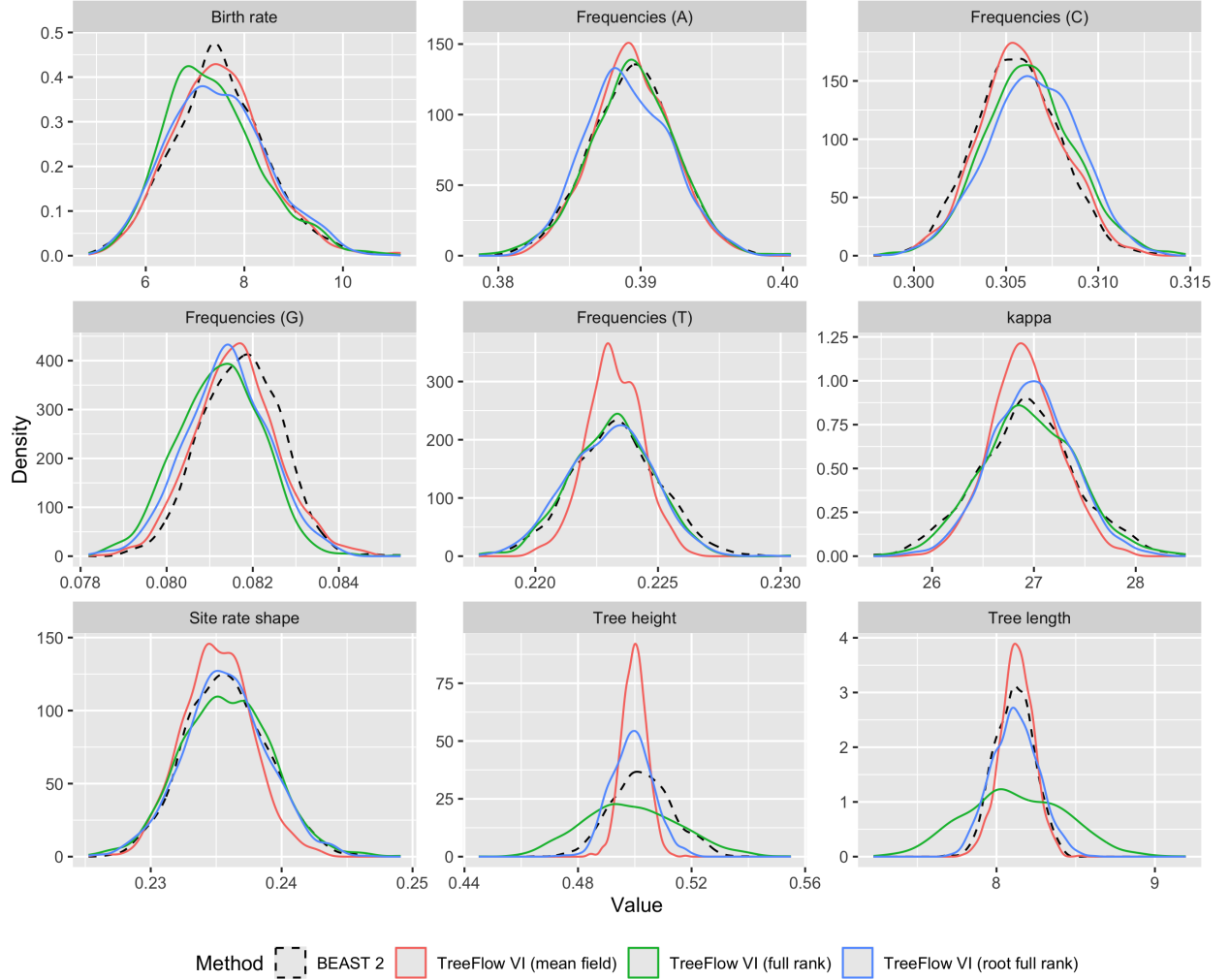


Figure S3: Posterior marginals for the carnivores dataset under each of TreeFlow’s three variational approximation families (solid lines), against the BEAST 2 reference posterior (dashed). One variational run per family.

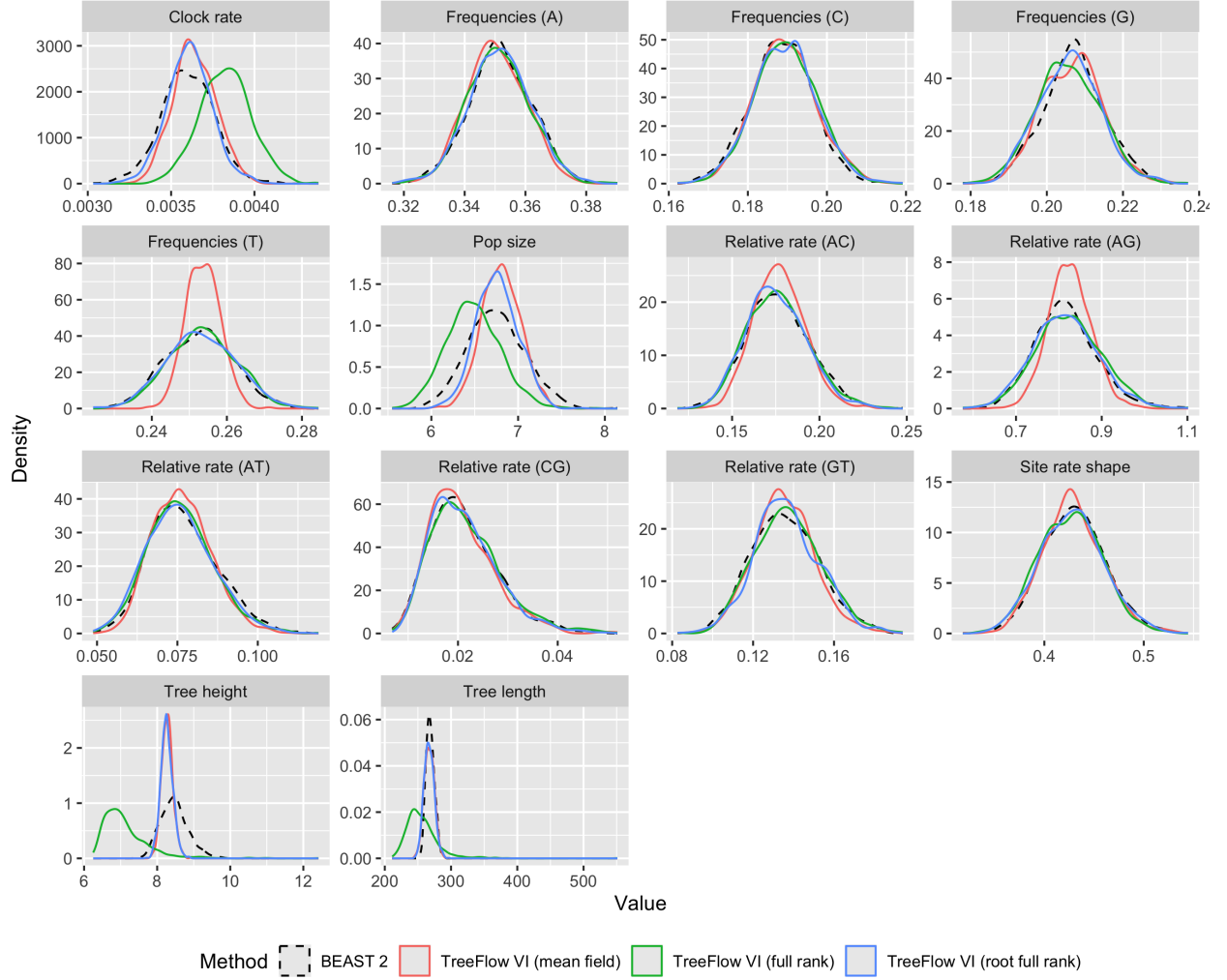


Figure S4: Posterior marginals for the influenza (H3N2) dataset under each of TreeFlow's three variational approximation families (solid lines), against the BEAST 2 reference posterior (dashed). One variational run per family.